Performance and Accuracy Tradeoffs of Low Precision KV Cache Quantization

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Motivation:

This project will study key-value (KV) cache quantization for Qwen3.5-9B on one NVIDIA H100 GPU, examining when reduced memory traffic outweighs reconstruction overhead and accuracy loss during inference [1, 2].

Approach:

With BF16 weights, I will compare BF16, FP8 E4M3/E5M2, INT8, INT4, FP4 E2M1, and INT2 KV caches, with KIVI informing low-bit integer quantization [4]. A compatibility pilot will precede packed KV/attention microbenchmarks for all formats and vLLM generation tests for integrated formats. FP4 tests on H100 will include software reconstruction costs [5]. Full-model tests will cross 2K/4K/8K/16K-token prompts (K = 1,024) with concurrency 1/4/8/16, generating exactly 256 tokens per request. Each configuration will use warmup and three 128-request runs with matched inputs. Under a common cache memory budget, I will extend concurrency at 8K/16K, recording queueing and preemption. Thinking, prefix caching, and speculative decoding will be disabled. Software and scheduler settings will be fixed within each backend; backend variants will be reported separately. Scaling, grouping, residual buffers, and compute precision will be documented, with disjoint calibration data. This hybrid model's recurrent states will retain their original precision and be accounted separately [3]. Integrated formats will undergo 100 retrieval and 100 labeled QA tests with deterministic decoding and natural stopping.

Results & Significance

I will measure cache bytes with metadata, total GPU memory, resident capacity, output throughput, first-token and per-token latency, and median/P95 request latency. Profiling will separate quantization, unpacking, attention, and other costs. Reconstruction and attention output errors will complement exact-match/F1 scores. Plots will show memory, throughput, and quality tradeoffs; Amdahl's law will explain speedup limits. Simulations will assess numerical error only; findings will remain workload-specific.

Future Work

FP2 exploration requires a defined public format and implementation. Deliverables comprise reproducible experiments, a final report, and a short video.

Bibliography

[1] W. Kwon et al., *Efficient Memory Management for Large Language Model Serving with PagedAttention*, SOSP, 2023.

[2] J. Kübler et al., *The State of FP8 KV-Cache and Attention Quantization in vLLM*, vLLM, 2026.

[3] Qwen Team, *Qwen3.5-9B Model Card*, Hugging Face, 2026.

[4] Z. Liu et al., *KIVI: A Tuning-Free Asymmetric 2bit Quantization for KV Cache*, 2024.

[5] E. Alvarez et al., *Introducing NVFP4 for Efficient and Accurate Low-Precision Inference*, NVIDIA Technical Blog, 2025.